

Study Quick Guide

1. Overview

- This study includes 4 conditions and 3 tasks.
- Conditions: Macro - Near Head, Macro - Side Of Body, Micro - Near Head, Micro - Side Of Body
- Tasks: Placement, Rotation, Scaling
- The condition order may vary depending on the Participant ID.
- In both Macro and Micro, submission is always triple tap for every task.

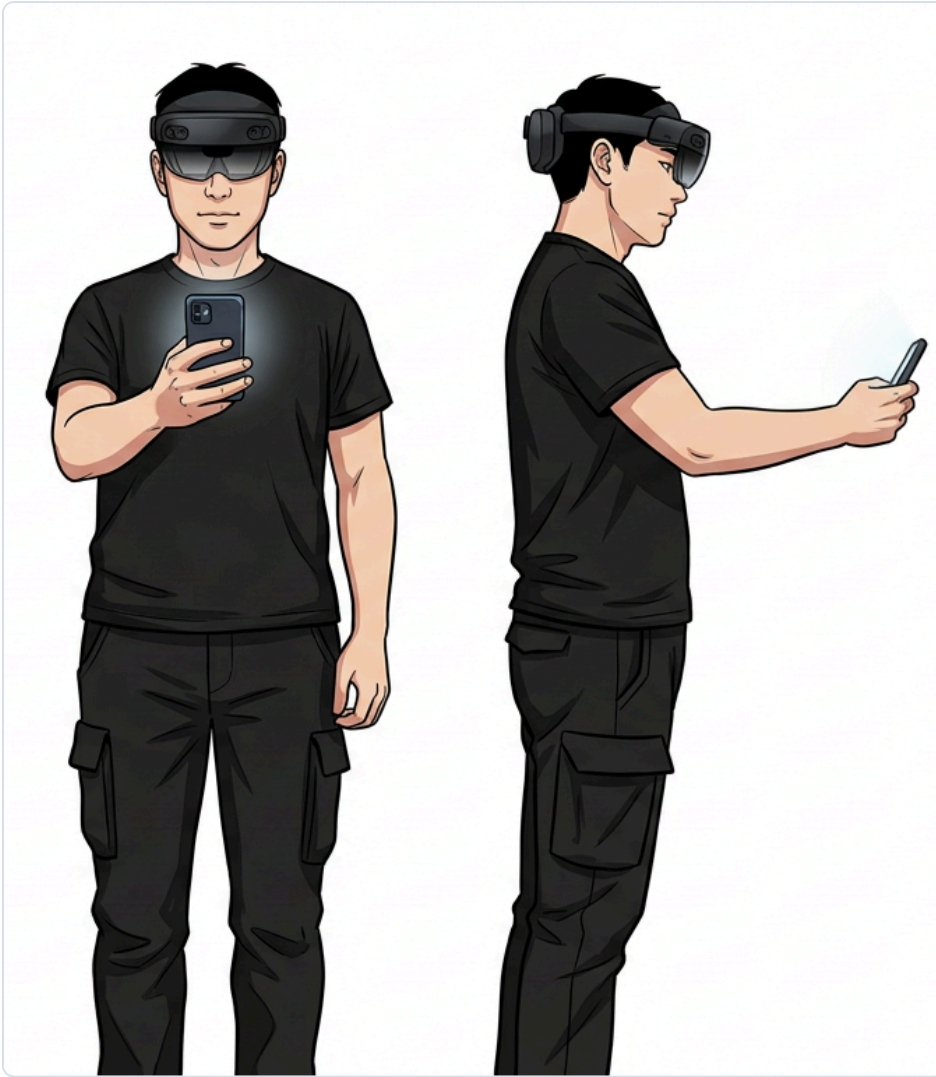
2. Procedure

1. On the start screen, enter your assigned Participant ID.
2. If needed, use Edit to change it, then press Continue.
3. In the runtime scene, when you see Scan the QR code to begin., scan the HoloLens workspace QR code.
4. At the start of each condition, open or restart the phone app, scan the QR code with the phone, and wait for the HoloLens to say Phone QR detected.
5. Follow the posture instruction and example image, then hold that posture briefly.
6. After the countdown, the task starts automatically. There is no separate Start button.
7. A short practice may appear before each task. Practice is not logged.
8. Match the target, then triple tap when you are ready to submit.
9. After one condition ends, the next condition starts automatically.

3. Condition Summary

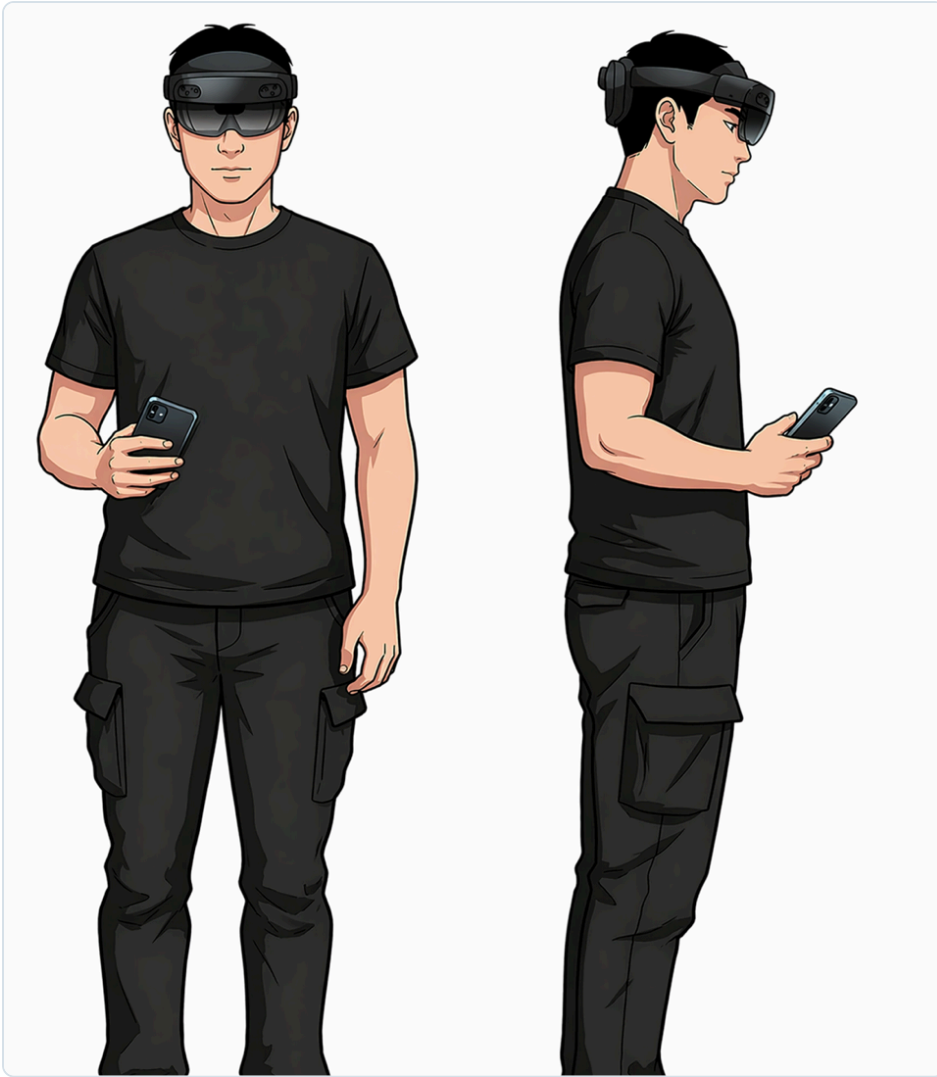
Macro - Near Head

- Use larger arm and hand movements.
- Keep your hand near your head.



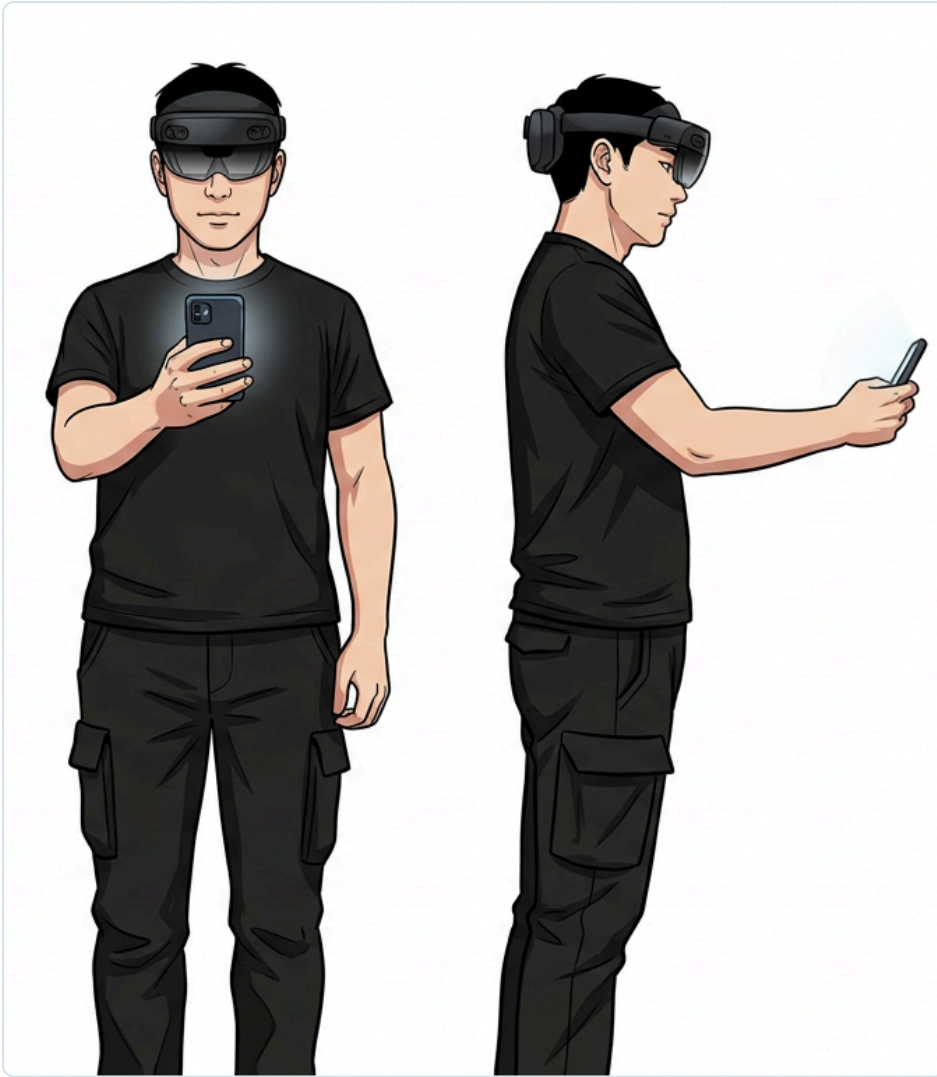
Macro - Side Of Body

- Use larger arm and hand movements.
- Keep your hand at the side of your body.
- In this condition, the input is remapped.
- Up/down hand movement is remapped to forward/back movement, and forward/back movement is remapped to up/down movement.



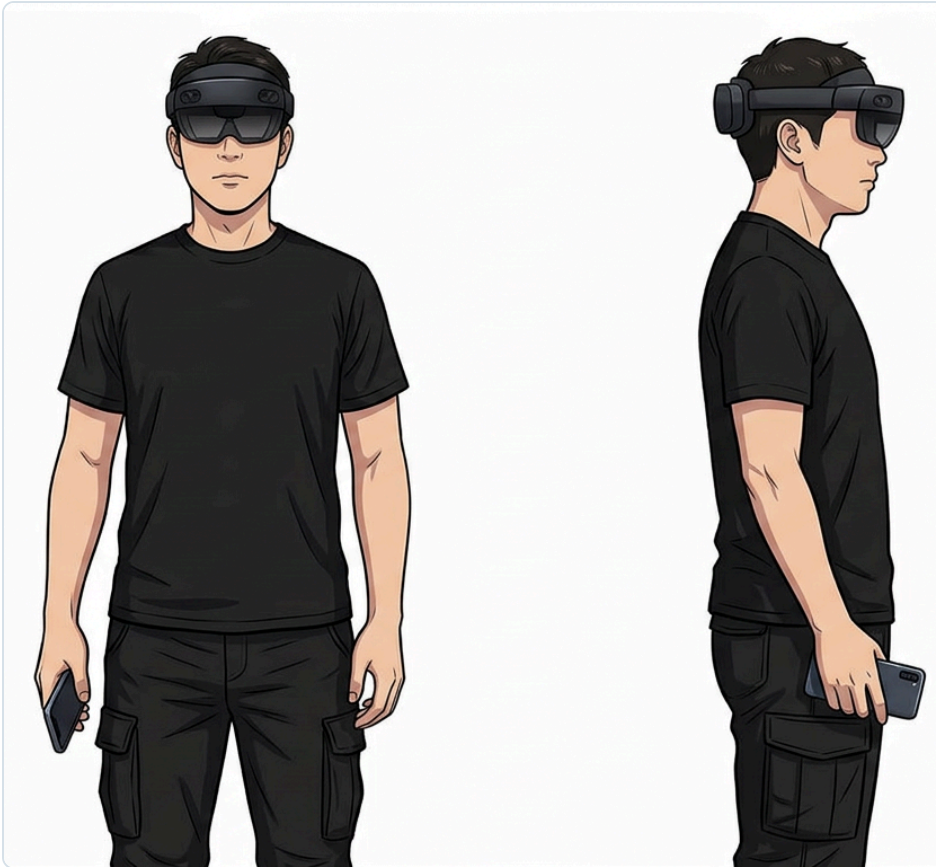
Micro - Near Head

- Keep your hand near your head.
- Use phone screen swipes for fine control.



Micro - Side Of Body

- Keep your hand at the side of your body.
- Use phone screen swipes for fine control.



4. Task Summary

Placement

- Move the tool to the target position.
- In **Micro**, swipe on the phone to move the tool left/right and up/down.
- If needed, **double tap** to switch the movement plane for forward/back adjustment.
- When ready, submit with **triple tap**.

Rotation

- Match the target orientation.
- In **Micro**, left/right swipes control yaw, and up/down swipes control roll.
- If needed, **double tap** to switch the vertical swipe axis between **roll** and **pitch**.
- When ready, submit with **triple tap**.

Scaling

- Match the target size.
- In **Micro**, up/down swipes make the tool larger or smaller.
- When ready, submit with **triple tap**.

5. Notes

- In **Macro**, you control the task mainly with arm and hand movement.
- In **Micro**, you keep the instructed hand posture and use phone swipes for fine adjustments.
- If the phone app is restarted for a new condition, scan the QR code with the phone again before matching the posture.
- Try not to change posture too much while the instruction image and countdown are shown.
- If QR scanning fails, **Phone QR detected.** is not announced, or the display/input seems wrong, tell the facilitator immediately.